

Meeting Summaries of December 18

Time: 1:00–1:45 PM(PDT)

Attendees: Prof. Jan, Dr. Javed, Dr. Yue and Feng

Topic: Feng's report on RDA and LDA results; overview of complete workflow. Plans and schedules determined for and after the winter break.

Summary of Discussion

1. Dr. Yue invited Prof. Jan and Dr. Javed for the speakings at Data Science seminars at TRU next year, the tentative dates are around **February 10 and 17**.
2. Feng reported the inputs, brief methodology and results of RDA, and the performance of LDA, summarizing the purposes of taking the two methods for habitat-community linkage analysis.
3. Feng gave an overview of the important steps and parameters through the complete workflow up to LDA. They are tentatively named *global settings*, including:
 - (a) **Transformation on PC scores:** Assumptions and standardization applied to pollution-related PC scores (implemented in 02 notebook).
 - (b) **Selection of reference sites:** Approach to select the least polluted $p\%$ of sites (02).
 - (c) **Transformation and interpretation of species abundance data:** Transformation applied to abundance data for dissimilarity among sites in terms of species and for clustering (03).
 - (d) **Number of clusters (k):** Number of clusters to partition reference sites (03).
 - (e) **Construction of RDA axes:** Regression model choice for predicting the constrained species matrix (response values) for RDA axes (04).
 - (f) **Predictor selection for LDA:** Habitat variables selected as LDA predictors that comprise the discriminant splits to divide sites into clusters.(04).
4. Prof. Jan provided *feedbacks* on the LDA operation and results:
 - (a) How the **confusion matrix** is for the classification of four cluster. That is to see what specific misclassifications happened among which clusters (there are priorities and explanations behind these misclassifications).
 - (b) Is it possible to **skip the poor bifurcation** for two resulted sites that are difficult to separate? Additionally, the splits from LDA that define the predicted groups should be revealed and interpreted and replacing the bifurcation with a fusion of these sites might give a better classificatoion performance.
 - (c) (*From an email before meeting, about the clustering work last week.*)
 - i. The selection of k cluster needs to be **shown in the dendrogram and be commented on their properties**, species compositions, habitat characteristics, etc.
 - ii. It is good to **check** Feng's results and comments with what Jian identified, and keep results balanced between statistical performance and ecological and biological sense.

- iii. Also, the **group integrity needs to be validated** after the groups been chosen, which has nothing to do with "how correct the clusters were made through the process" but with "how well the selected groups reveal ecological and biological senses."
5. Dr. Javed suggested to try different cross-validation methods to evaluate the LDA performance, Leave-One-Out CV is not sufficient.
6. Dr. Javed advised Feng to finish the complete computation first, finish the modelling and computation in December, which is good for the winter break plan.
7. Dr. Javed suggested Feng to prepare for the *Conference of Statistical Society of Canada*, the submission needs at least **a title and an abstract**, Feng should leave **at least two weeks** for supervisors to review the draft before it is submitted to.
8. Feng asked about the PhD application, there will be **4 submissions by December end**, which needs reference letters by **around the middle of January**.
9. The schedule of winter meeting time was decided, meetings will resume at **2:30 (PDT) on January 8th 2026**, then scheduled every **Thursdays at 2:30 PM (PDT)** afterwards.

Action Items

- Feng will finish the complete modelling and computation in December, the last two stages for ordination and regression.
- Feng will fix the missings based on the suggestions from Prof. Jan and Dr. Javed, organize the workflow and findings clearer.
- Feng will submit the rest 4 applications in December.
- Feng will check the information and prepare for the Conference of Statistical Society of Canada in May 2026.

Follow-Up

Temporary meeting time: 2:30 (PDT) on January 8th

Supportive materials

- A meeting recording has been restored on Feng's laptop.